FULL SIMULATION SET 04

UPSC Civil Services (Main) Simulation GS-III

DETAILED SOLUTIONS / ANSWER KEY | Time: 3 Hours | Maximum Marks: 250

Q1. Explain the inverse bond price-yield relationship and show its importance for monetary transmission.

[10 marks | 150 words]

Model answer (130 words; practice model, not official): A fixed-coupon bond promises contractual cash flows. Its price equals their present value discounted at the required yield. When market yield rises, the higher discount rate lowers present value; when yield falls, price rises. A below-par bond can therefore have a current yield above its coupon rate, while yield to maturity also includes redemption gain or loss. Duration converts this principle into risk measurement: longer-duration bonds show larger price changes for a small yield movement. Convexity makes duration only an approximation. The mechanism matters for monetary transmission. Expected short rates and liquidity alter the sovereign yield curve, which influences bank funding, corporate-bond pricing and investment valuation. Yet inflation expectations, term premium, bond supply, credit spreads and global conditions may offset policy. The curve is both transmission channel and conditional signal.

Q2. Why can MSP provide only partial protection against low farm income?

[10 marks | 150 words]

Model answer (122 words; practice model, not official): MSP is a pre-season administered price for specified crops, not a universal income guarantee. CACP recommends and CCEA decides. The policy benchmark is at least 1.5 times the all-India weighted average A2+FL cost. Protection remains partial because announcement differs from procurement. A farmer must offer eligible produce, meet Fair Average Quality, reach an operating centre and sell during the procurement period. Purchase is concentrated in paddy, wheat and established States, while many producers sell locally. Farms with higher costs may not realise the published average margin. Income also depends on yield, area, input prices, crop loss and weather. MSP can reduce downside price risk and guide sowing, but broader income security requires productivity, diversification, insurance, market access and appropriate transfers.

Q3. Why is Public Private Partnership (PPP) required in infrastructural projects ? Examine the role of PPP model in the redevelopment of Railway Stations in India.

[10 marks | 150 words]

Model answer (145 words; practice model, not official): PPPs can combine public service objectives with private finance, design, construction, commercial development and lifecycle management where outputs and risks are contractible. Railway-station redevelopment requires integration of passenger circulation, safety, accessibility, platform interfaces, urban transport and commercial space. A PPP may use station-area revenue to support modernisation, allocate construction and maintenance duties, and impose availability and service standards. It can reduce immediate public capital needs and encourage whole-life design. Yet station demand, land title, heritage, security, railway operations and tariff authority cannot be shifted mechanically. Overoptimistic commercial revenue, weak urban coordination or unclear termination payments can trigger renegotiation. The public authority should complete site and demand preparation, protect core railway functions, competitively bid transparent rights, monitor accessibility and service quality, disclose contingent liabilities and retain remedies. PPP is useful when it delivers better lifecycle value than public procurement, not merely because a private developer participates.

Q4. Explain why MSME policy should support graduation rather than permanent smallness.

[10 marks | 150 words]

Model answer (120 words; practice model, not official): MSMEs broaden entrepreneurship, regional industry and supplier networks, but policy should promote productive graduation rather than preserve smallness. Economic Survey 2025-26 reports their large manufacturing, export and GDP footprint. Yet Udyam registration is an administrative identity, not proof of active production or productivity. Micro firms face different constraints: collateral, unpaid invoices, technology, standards, skills and market access. An effective strategy therefore sequences Udyam, suitable credit or TReDS, payment enforcement, shared cluster facilities, ZED quality systems and procurement access. Transition support should reduce the cost of crossing size thresholds. The test is not the number of registered enterprises. It is survival, formal jobs, wages, productivity, supplier upgrading and movement into competitive medium firms, with targeted support for women and disadvantaged entrepreneurs.

Q5. Explain how input-tax credit and destination-based IGST jointly support a common market.

[10 marks | 150 words]

Model answer (124 words; practice model, not official): GST combines input-tax credit with destination-based IGST. An eligible registered person offsets tax paid on business inputs against output liability, so net payment approximates tax on value added. This limits cascading and supports invoice reporting, subject to eligibility and documentation. For inter-State supplies, Article 269A authorises Union levy and collection of IGST, while credit and settlement rules transfer revenue toward the destination. Supplies can cross State borders without breaking credit at each border or assigning tax to the producing State. Together, these mechanisms support a common market through neutrality and destination assignment. However, blocked credits, exempt links, place-of-supply disputes, delayed refunds and fraud can weaken that result. The architecture reduces fiscal barriers; it does not guarantee frictionless trade without sound administration and prompt settlement.

Q6. Differentiate primary and secondary succession using the soil and biological-legacy test.

[10 marks | 150 words]

Model answer (92 words; practice model, not official): 1. Primary succession begins on a substrate without a developed soil, such as newly exposed rock; soil formation and biological colonisation are part of the process, but no universal completion timeline is asserted. 2. Secondary succession follows disturbance where soil and often propagules or a seed bank remain; it commonly proceeds faster than primary succession because the substrate and biological legacies differ. 3. Pioneer organisms tolerate the starting substrate and begin modifying it; lichens are the classic routed example for exposed surfaces without soil, but pioneer identity depends on the actual substrate.

Q7. Explain why a satellite launch is not the same as a satellite service.

[10 marks | 150 words]

Model answer (148 words; practice model, not official): 1. Earth-observation, communication, meteorological and navigation satellites are classified by payload and service, not merely by orbit or spacecraft name. 2. A geostationary orbit supports continuous view of a region, while polar and sun-synchronous low orbits support repeated Earth coverage; orbit enables but does not uniquely determine a satellite's service. 3. A satellite service requires a space segment, ground control and data-processing segment, and user receivers or departments; a spacecraft in orbit is not the same as a delivered service. 4. Earth-observation sensors collect reflected or emitted energy, ground systems process it, and users convert products into mapping, agriculture, water, disaster and planning decisions. 5. A communication transponder receives, shifts, amplifies and retransmits signals; orbital hardware becomes a service only through terrestrial networks and user terminals. 6. Meteorological satellites combine imaging, sounding, data relay and warning support; a weather picture is only one output of the observation system.

Q8. Explain why LWE is not merely a law-and-order problem.

[10 marks | 150 words]

Model answer (122 words; practice model, not official): 1. Left-Wing Extremism is a Maoist armed challenge aimed at capturing state power through protracted people's war, not a secessionist claim for a separate sovereign State. 2. Land alienation, weak recognition of traditional forest rights, acquisition without adequate compensation and disruption of jal-jangal-zameen relationships are core enabling grievances. 3. Poverty, unemployment, infrastructure deficit, administrative absence, weak justice, social exclusion and alienation create exploitable vulnerabilities but do not mechanically cause violence. 4. LWE intensity historically overlapped mineral-rich areas because infrastructure, mining and contracting activity created extortion opportunities; resource presence alone does not explain recruitment. 5. available evidence indicates that Maoists can obstruct roads, electricity, water and administration because continued underdevelopment sustains territorial influence; the movement may maintain the condition it claims to oppose.

Q9. Distinguish the disaster-warning roles of IMD, CWC, INCOIS, ISRO and NDMA.

[10 marks | 150 words]

Model answer (103 words; practice model, not official): 1. IMD monitors and forecasts weather hazards and issues official meteorological and cyclone warnings, including impact information where its service supports it. 2. CWC monitors river conditions and issues flood forecasts and warnings to administrations, project authorities, States and relevant Central agencies. 3. INCOIS operates the Indian Tsunami Early Warning Centre, integrating earthquake analysis, pre-run scenarios and sea-level observations for tsunami bulletins. 4. ISRO and NRSC provide Earth-observation, remote-sensing, GIS and decision-support products for preparedness, event assessment and recovery planning. 5. NDMA supports national alert integration and guidance, while State and local authorities translate authoritative warnings into evacuation, shelter, route and public-action decisions.

Q10. Why can a bond generate loss even when its issuer makes every promised payment?

[10 marks | 150 words]

Model answer (132 words; practice model, not official): A bond's contractual performance does not fix its resale value or realised return. When market yields rise, the present value of fixed coupons and principal falls, producing a mark-to-market loss. Longer duration increases this sensitivity. A widening liquidity premium can force a larger price concession even without deterioration in issuer solvency. Reinvestment risk works differently: coupons received during the holding period may be reinvested at lower rates. Inflation can erode the purchasing power of nominal payments. Callable bonds can be redeemed when rates fall, removing upside and forcing reinvestment. Foreign-currency exposure can change the investor's home-currency return. Thus credit risk is only one component. An investor should distinguish default, interest-rate, duration, liquidity, reinvestment, inflation, optionality and currency risks. Fixed income describes the cash-flow rule, not a guaranteed market price or real return.

Q11. Explain the national-income aggregate ladder and the treatment of cross-border remittances.

[15 marks | 250 words]

Model answer (230 words; practice model, not official): National-income aggregates change one accounting boundary at a time. GVA at basic prices plus net taxes on products gives GDP at market prices. GDP measures production within domestic territory. Adding primary income receivable from the rest of the world and subtracting primary income payable gives GNI, commonly called GNP in older texts. Deducting consumption of fixed capital gives NDP from GDP and NNI from GNI. The next bridge concerns disposable resources rather than production income. Gross national disposable income equals GNI plus net current transfers from abroad; the net version deducts CFC. This is why all remittances cannot be added to GDP or treated as net factor income. Compensation for labour and property income are primary income when residence rules make them cross-border. A personal transfer sent without a corresponding labour, capital or service claim is secondary income/current transfer and affects disposable income. Named evidence comes from MoSPI's National Accounts Statistics 2026, which publishes GDP, national income, per-capita income and external transactions as distinct statements. The analytical gain is precision: GDP answers where production occurred; GNI answers whose primary income accrued; disposable income answers what resources residents can consume or save. The qualification is measurement. Residence is not citizenship, transfer classifications depend on transaction characteristics, and CFC is estimated. A sound answer therefore writes every bridge explicitly and avoids the textbook shortcut 'income from abroad' unless it is decomposed.

Q12. Why did land reforms achieve deeper results in some Indian States than in others? Analyse in 250 words.

[15 marks | 250 words]

Model answer (202 words; practice model, not official): Land reform is principally implemented through State legislation, so a common national objective produced different field outcomes. Success required more than progressive statutory language. First, political coalitions had to support cultivators against concentrated landed interests. West Bengal's Operation Barga combined sharecropper recording with mobilisation, making tenancy protection visible and enforceable. Kerala's comparatively deep tenancy and ceiling reforms similarly reflected sustained political backing. The early Big Landed Estates Abolition Act, 1950 in Jammu and Kashmir shows how timing and a distinct political setting could permit sharp restructuring. Second, administrative information mattered. Updated records, cadastral maps and local verification identified tenants, holdings and surplus. Weak records enabled benami transfers, anticipatory partitions and exclusion of actual cultivators. Third, implementation required a chain from declaration to adjudication, possession, allotment and mutation; delay at any stage reduced the reform. Fourth, beneficiary organisation improved reporting and bargaining power. Fifth, complementary credit, irrigation, extension and market access determined whether redistributed or secured land became a viable productive asset. Yet these examples should not be romanticised. Tenure security did not eliminate fragmentation, tiny holdings, labour underemployment or later agrarian stress. Regional success was therefore the product of aligned law, politics, records, administration and support services, not any single legal instrument.

Q13. Evaluate coal's changing role in India's energy security and transition.

[15 marks | 250 words]

Model answer (174 words; practice model, not official): Coal remains central because existing plants, domestic mines and rail-linked infrastructure supply large volumes of dispatchable electricity and industrial fuel. The Ministry of Coal reports 1,047.523 MT production in FY 2024-25, yet FY 2025-26 imports were 246.37 MT, including 66.33 MT coking coal. Security depends on grade and delivered logistics, not aggregate production alone. Revised SHAKTI, approved 7 May 2025, addresses power-sector linkages, while commercial mine auctions concern another allocation route. Coal also imposes air, water, ash, land and carbon costs. As solar and wind expand, thermal plants may run fewer hours but need better ramping and reserve performance; lower PLF can raise per-unit fixed-cost recovery. Abrupt retirement can strand PPAs and debt and concentrate job, royalty and local-economy losses. Policy should prioritise efficient units, logistics, pollution compliance, flexible operation, transparent retirement criteria and closure funds while scaling grids, storage and demand response. A just transition needs worker protection, land restoration and regional diversification. Coal's role should decline through planned substitution of energy and reliability services, not a capacity-only pledge.

Q14. Evaluate the achievements and limitations of India's 1991 industrial reforms.

[15 marks | 250 words]

Model answer (220 words; practice model, not official): The 1991 reforms responded to a balance-of-payments crisis but went beyond emergency stabilisation. The New Industrial Policy dismantled most industrial licensing, reduced public-sector reservation, removed MRTP pre-entry restrictions and liberalised foreign investment and technology agreements. Trade, exchange-rate, financial and tax reforms complemented these changes. Delicensing improved entry and expansion incentives. Import and domestic competition pressured firms to modernise, while foreign capital and technology widened production choices. Efficient enterprises gained scale, consumers obtained greater variety and industrial allocation became less dependent on administrative discretion. The reforms also enabled deeper global-value-chain participation. Yet outcomes were uneven. Infrastructure, power, finance, skills and contract enforcement limited the response of many firms. Competitive restructuring displaced protected workers and increased skill premiums. Investment concentrated in regions with established cities, suppliers and logistics. Foreign investment did not automatically create domestic R&D or value addition. Weak competition or sector regulation could replace public monopoly with private market power. The reforms also left unfinished factor-market and State-capacity agendas. Land, logistics, urban governance, judicial delay, regulatory overlap and small-firm capability remain binding. Therefore 1991 should be judged as a transition from administrative allocation to market competition, not as a completed reform event. Second-generation reforms must combine infrastructure, skills, competition, social protection and capable regulation. Current industrial-policy resurgence should build measurable strategic capability without restoring open-ended protection or discretionary licensing.

Q15. Explain how Parliament authorises and scrutinises Union expenditure.

[15 marks | 250 words]

Model answer (219 words; practice model, not official): The Constitution separates fiscal disclosure, voting and legal withdrawal. Under Article 112, the President causes the Annual Financial Statement to be laid before both Houses. It distinguishes charged expenditure from other expenditure and separates revenue-account spending. Under Article 113, charged expenditure may be discussed but is not submitted to vote. Other expenditure is presented ministry-wise as Demands for Grants to the Lok Sabha, on the President's recommendation. Members can use policy, economy and token cut motions to question policy, amount or a specific grievance. The guillotine, however, may dispose of undiscussed demands when allotted time expires. After voting, Article 114 requires an Appropriation Bill covering voted grants and charged sums. No money can ordinarily be withdrawn from the Consolidated Fund without appropriation. The Finance Bill separately gives effect to taxation proposals. Article 115 provides supplementary, additional and excess grants when provision is inadequate, a new service arises or expenditure exceeds authority. Article 116 permits Vote on Account, vote of credit and exceptional grants. Control continues after enactment. Ministries execute expenditure; the Controller General of Accounts compiles accounts; the Comptroller and Auditor General audits; and parliamentary committees examine compliance and performance. Thus parliamentary control is not confined to Budget Day. Its effectiveness depends on realistic estimates, adequate debate, committee scrutiny, transparent supplementary demands, timely accounts and follow-up on audit findings.

Q16. Explain radiative forcing, feedbacks and gas-specific climate metrics.

[15 marks | 250 words]

Model answer (173 words; practice model, not official): 1. Radiative forcing is an imposed change to the climate system's energy balance, while a feedback is a response triggered by climate change that amplifies or dampens the initial change. 2. A positive climate feedback amplifies an initial change and a negative feedback dampens it; positive and negative describe direction, not whether the outcome is desirable. 3. Water vapour is a major greenhouse gas and generally operates as a feedback in contemporary warming because atmospheric moisture responds strongly to temperature; it is not a substitute for tracing the initiating anthropogenic forcing. 4. Many aerosols exert a cooling influence through scattering and cloud effects, while black carbon absorbs radiation and can reduce snow or ice albedo; 'aerosol' does not imply one universal forcing sign. 5. Global Warming Potential compares integrated radiative influence relative to carbon dioxide over a specified time horizon; a GWP claim without its horizon is incomplete. 6. Heat-trapping potency per unit mass and atmospheric persistence are different properties; a short-lived strong forcer and a long-lived cumulative gas require different mitigation reasoning.

Q17. Explain the D-T fusion chain and its tritium and materials constraints.

[15 marks | 250 words]

Model answer (183 words; practice model, not official): 1. The deuterium-tritium reaction produces helium-4, a neutron and 17.6 MeV; the neutron carries most energy outward while the alpha particle supports plasma self-heating. 2. Fusion fuel must become an ultra-hot ionised plasma to overcome electrostatic repulsion; a quoted temperature is a plasma condition, not proof of confinement, gain or plant readiness. 3. The Lawson triple product links plasma density, temperature and energy-confinement time; improving one variable alone does not satisfy the fusion condition. 4. Charged alpha particles can remain magnetically confined and heat the plasma, whereas 14.1 MeV neutrons escape magnetic confinement, heat the blanket and damage or activate structures. 5. Tritium is scarce and radioactive, so a future D-T plant must breed it from lithium-bearing blankets with losses and decay considered; ITER tests concepts rather than proving a closed commercial fuel cycle. 6. High-energy neutrons cause displacement damage, helium production and activation, while plasma-facing components face intense heat loads; material survival is separate from plasma confinement. 7. A divertor extracts heat, impurities and helium ash from the plasma edge; successful core confinement does not solve exhaust and component-lifetime constraints.

Q18. Discuss why Centre-State coordination is a structural requirement in internal-security governance.

[15 marks | 250 words]

Model answer (157 words; practice model, not official): 1. The Ministry of Home Affairs is the Union ministry responsible for internal security, while the Ministry of Defence owns external defence; central assistance does not erase State policing competence. 2. Public order is State List Entry 1 and police is State List Entry 2, making the State the primary day-to-day authority for internal order and investigation. 3. Union List Entry 2A permits deployment of an armed force of the Union in a State in aid of the civil power; assistance supplements rather than replaces the State authority. 4. Article 355 places a duty on the Union to protect every State against external aggression and internal disturbance and to ensure constitutional government. 5. Because policing is primarily State-owned while intelligence, border guarding and several central capabilities are Union-linked, Centre-State coordination is a structural constitutional requirement. 6. A legal power is not operational capability: trained personnel, forensics, court time, lawful procedure and inter-agency trust remain separate implementation conditions.

Q19. Analyse the warning-to-evacuation and cyclone-shelter preparedness chain.

[15 marks | 250 words]

Model answer (149 words; practice model, not official): 1. IMD monitoring and forecasts communicate track, intensity, rainfall, wind, sea condition, storm-surge and likely impact information with uncertainty; a forecast is not a deterministic outcome. 2. Green, Yellow, Orange and Red communicate action levels, while Pre-Cyclone Watch, Cyclone Alert, Cyclone Warning and Post-Landfall Outlook form a separate lead-time bulletin sequence. 3. Authorities convert official forecasts into area-specific evacuation, sheltering, route control, transport, livestock and asset-protection decisions, with priority assistance for people unable to self-evacuate. 4. Multi-purpose cyclone shelters require safe siting, all-weather access, water, sanitation, backup power, accessibility, protection, management and maintenance; construction alone does not prove readiness. 5. Warnings require trusted multilingual relay, drills, local volunteers, accessible transport, route familiarity and shelter management; warning issuance does not demonstrate household action. 6. A forecast, colour code, alert, shelter, embankment, ecosystem project or deployment proves an input; timely evacuation, service continuity, maintained assets and reduced loss require separate verification.

Q20. Explain the major sources of inflation in India and show how headline, core, food and fuel measures aid diagnosis.

[15 marks | 250 words]

Model answer (220 words; practice model, not official): Inflation is a sustained rise in the general price level, but its policy meaning depends on source, breadth and persistence. Demand-pull inflation arises when aggregate spending exceeds available capacity; fiscal stimulus, rapid credit growth or strong purchasing power can close the output gap. Cost-push inflation follows higher wages, taxes, freight or input prices. Supply shocks arise from crop loss, storage and transport failures or administered-price changes. Imported inflation transmits global crude, edible-oil, fertiliser and freight costs through the exchange rate and domestic supply chains. Headline CPI records the full consumer basket and therefore captures the prices households actually face. CFPI isolates food pressure, while fuel components reveal energy pass-through. Core inflation, conventionally headline excluding food and fuel, helps assess underlying persistence; however, it is an analytical construct and can itself be distorted by unusual items. Economic Survey 2025-26, for example, examined precious metals when interpreting core pressure. The categories interact. A vegetable shock is initially relative and supply-led, but repeated shocks can alter wages, margins and expectations. Rural inflation may also be more volatile because food has a larger budget role. Policy must therefore combine RBI action against broad demand and second-round effects with buffers, logistics, calibrated trade, tax measures and productivity improvements. The qualification is crucial: no component is dispensable, and no single index reproduces every household's experienced inflation.

END OF DETAILED SOLUTIONS